Pydimarri Venkata Naga Sindhu

me23btech11051@iith.ac.in | pydimarri.sindhu123@gmail.com | +91 7013272771 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.76
Sri Chaitanya College, Vijayawada	Class XII (SSC)	2023	98.80%
Dr. KKR Gowtham Concept School, Gudivada	Class X (SSC)	2021	99.66%

TECHNICAL SKILLS

- **Languages :-** C, C++, Python
- **Python Libraries :-**NumPy, Pandas, Matplotlib, SciPy
- **Data Science & AI :-** Machine Learning(ML), Neural Networks, Deep Learning Fundamentals
- **Database Systems :-** MySQL
- **Development Tools :-** VS Code, GitHub, Google Colab

ENGINEERING SOFTWARE

CAD :-Solid Edge, SolidWorks, Fusion 360

FEA :-Abaqus

CFD :- ANSYS Fluent

Languages :- C, C++, Python

RELEVANT COURSES

Engineering Drawing

Thermodynamics, Heat and Mass Transfer, Power and Refrigeration Systems

Engineering Mechanics, Solid and Fluid Mechanics

Manufacturing Science - 1&2

Kinematics and Dynamics of Mechanisms, Design of Machine Elements

Finite Element Method(FEM), Computational Fluid Dynamics(CFD)

Modeling and Simulation

PROJECTS

Jumping Sparrow Toy - Kinematic & Mechanism Analysis

- Disassembled and studied the internal mechanism, identifying gears, prismatic joints, springs, and hinged linkages used for motion generation.
- Developed the kinematic diagram and performed mobility analysis using the Kutzbach criterion for both original and modified configurations.
- Formulated loop-closure equations and carried out displacement, velocity, and acceleration analysis to characterize dynamic behavior.
- Used analytical results to evaluate motion characteristics, constraints, and design feasibility of the mechanical system.

Finite Element Analysis of Mechanical Components (Abaqus)

- Performed structural, thermal, and buckling analyses using Abaqus on 1D, 2D, and 3D mechanical models.
- Applied realistic loads, boundary conditions, and material properties to evaluate stress distribution, deformation, and thermal response.
- Conducted mesh convergence and stress concentration studies on notched components and plates with holes, using symmetry to improve computational efficiency.
- Used FEA results to validate mechanical designs and identify critical regions, strengthening analytical problem-solving skills.

Hydrogen Combustor - Thermal & Fluid Dynamics Analysis (CFD)

- Modeled reacting hydrogen-air flow inside a combustor using ANSYS Fluent to study thermal-fluid interactions under varying operating conditions.
- Performed parametric analysis on equivalence ratio (controlled via inlet velocity) and swirl-air temperature to evaluate combustion stability and thermal performance.
- Analyzed temperature fields, species concentration, velocity distribution, and fuel mass-fraction decay, identifying operating conditions for stable and efficient combustion.
- Interpreted CFD results to understand heat transfer, compressible flow behavior, and system performance limits, supporting informed engineering decisions.

Mechanical Test Frame for Instrumented Bar

- Designed and assembled a mechanical test frame for a Ø20 mm instrumented bar, focusing on structural integrity, alignment, and controlled axial motion.
- Selected and integrated standard off-the-shelf linear bearings (Misumi) into a custom assembly, designing precision housings based on supplier specifications.
- Designed an L-bracket to restrain axial movement under a 5 kN load and a pneumatic cylinder foot mount for a 10-bar air supply, considering load paths and boundary constraints.
- Completed full 3D assembly and production-ready 2D manufacturing drawings, emphasizing manufacturability, assembly, and mechanical design intent.

POSITIONS OF RESPONSIBILITY

Hospitality Coordinator, Milan 2024 (College Cultural Fest) - Managed guest coordination, logistics, and hospitality arrangements during the event.

EXTRACURRICULAR

- Member of **National Service Scheme (NSS)**, IIT Hyderabad

CERTIFICATIONS

- Professional Certification in **Placement Readiness & Career Impact Skills** by **Define Career**(Jun 2025)